

Supplementary Appendix

This appendix has been provided by the authors to give readers additional information about their work.

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Supplementary Appendix

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MAM, ACL, CDE, AMW, PAA, JRS, and DMS contributed to the design of the trial. MAM, CDE, PAA, SWA, JS, and SPS contributed to acquisition and/or analysis of data. MAM, ACL, CDE, AMW, PAA, SS, JS, JRS, MB, LGA, SPS, and DMS contributed to interpretation of data. All the authors contributed to drafting or critical revision of the manuscript, as well as reviewed and approved versions of the manuscript to be submitted for publication.

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Supplementary Methods

Global tau load

The acquired dynamic flortaucipir image was first corrected for motion, averaged and registered into a standard brain space. The Tau^{IQ} algorithm performed an image-based regression in order to model a corresponding Standardized Uptake Value Ratio image as the linear combination of two canonical images – the non-specific binding image and the tau carrying capacity image¹. The scalar factor of the carrying capacity is defined to be the global tau load. The Tau^{IQ} image analysis was conducted in Invicro, a Konica Minolta company.

Bayesian DPM

The Bayesian DPM was fit on the iADRS and other key secondary clinical outcomes to evaluate possible slowing of disease progression with treatment of LY3002813 relative to placebo. The primary purpose of the DPM is to estimate a quantity known as the disease progression ratio (DPR), which measures the proportion of disease progression in LY3002813-treated patients relative to placebo-treated patients. The key assumption of the DPM model is that it assumes the treatment effect of LY3002813 is proportional to placebo over the course of the study. The model includes diffuse priors on all parameters.

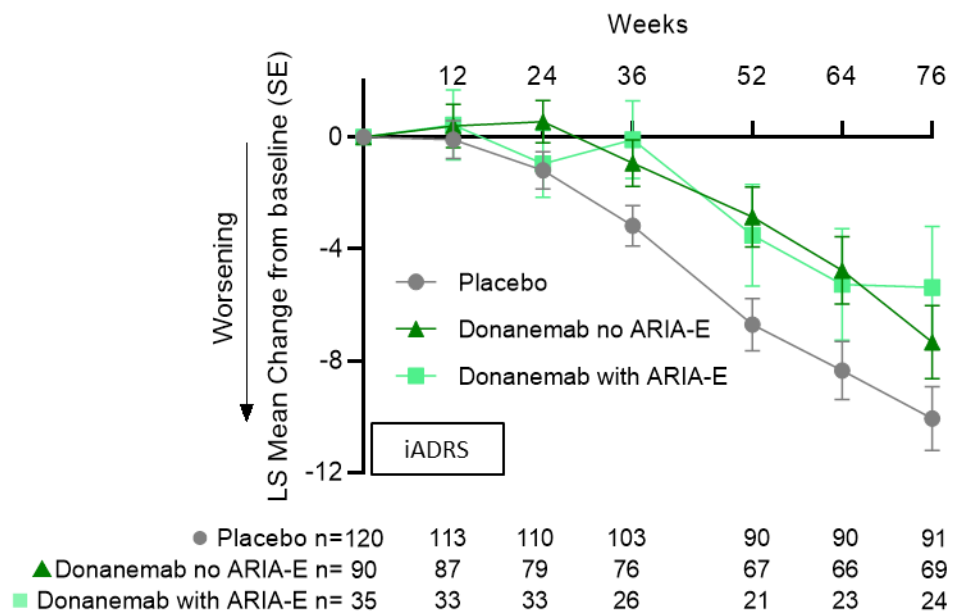
The DPM model is as follows:

$$Y_{ij} = \gamma_i + e^{\theta \tau_i} \sum_{v=0}^j \alpha_v + \varepsilon_{ij}$$

where Y_{ij} denotes the clinical outcome at visit j for patient i ; the clinical outcome score for a patient at baseline (prior to treatment) is Y_{i0} . The value γ_i ($i=1, 2, \dots, k$) represents a subject

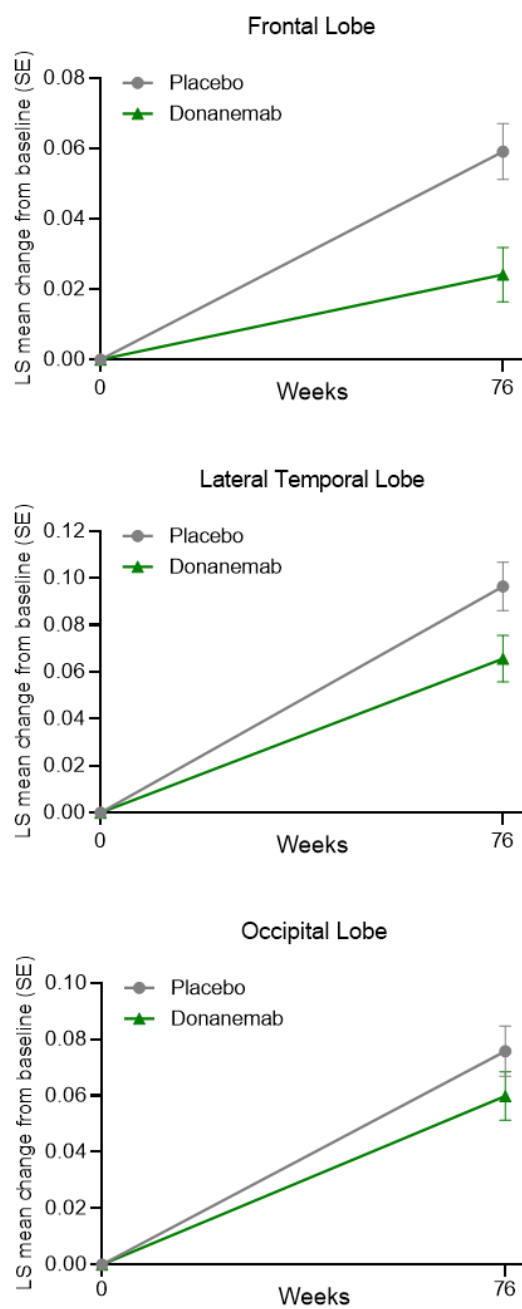
specific random effect. The parameter T_i denotes the treatment arm for patient i , where T_i has a value of 1 if a patient is randomized to LY3002813, and a value of 0 if the patient is randomized to placebo. The parameter α_v is the change in mean clinical outcome score for placebo from visit $v-1$ to v , and ε_{ij} is the error term. The DPR for LY3002813 relative to placebo is provided by the parameter e^θ . Additional covariates may be added to the model as needed. A DRR value less than 1 indicates the LY3002813 arm is slowing the disease progression relative to placebo. A DRR value greater than 1 indicates the LY3002813 arm is worsening the disease progression relative to placebo.

Figure S1: Results for the LS mean change from baseline to 76 weeks in iADRS score, by ARIA-E subgroup.



W=Week; iADRS=integrated Alzheimer's Disease Rating Scale; LS=Least Squares;
n=Number of participants; SE=Standard Error

Figure S2. Regional SUVr analyses of tau accumulation with cerebellar gray reference



LS=Least Squares; SE=Standard Error; SUVr=Standardized Uptake Value Ratio

Table S1. Reasons for screen failure during enrollment

Entered, N	1955
Screen failure, n (%)	1563 (79.9)
Reason for screen failure, n (%)	
Tau PET result	592 (30.3)
MMSE or lack of historical tau PET	347 (17.7)
CogState Brief Battery	334 (17.1)
Unreliable	59 (3.0)
Abnormal electrocardiogram	33 (1.7)
Study partner	22 (1.1)
Amyloid β PET result	17 (0.9)
Cancer history	16 (0.8)
Liver disease	15 (0.8)
Clinically important abnormality	14 (0.7)
Age	13 (0.7)
Unstable illness	12 (0.6)
Concomitant medication affect cognition	9 (0.5)
Presence of ARIA-E	9 (0.5)
Lack of consent	8 (0.4)
MRI contraindication	8 (0.4)
Neurological disease	7 (0.4)
Poor venous access	5 (0.3)
Concomitant medication AChEI	4 (0.2)
Concomitant medication prolong QTC	4 (0.2)

MHIS score ≥ 4	4 (0.2)
MRI	4 (0.2)
Contraindication to PET	3 (0.2)
Low creatine clearance	3 (0.2)
Psychiatric disorder	3 (0.2)
Alcohol disorder	2 (0.1)
Currently enrolled in clinical trial	2 (0.1)
Previous clinical trial too soon	2 (0.1)
Suicide risk	2 (0.1)
Concomitant medication CYP3A	1 (0.1)
Drug allergies	1 (0.1)
History of hepatitis C	1 (0.1)
History of long QT	1 (0.1)
History of hepatitis B	1 (0.1)
Inadequate premorbid literacy	1 (0.1)
Investigator personnel	1 (0.1)
Medical history of torsades de pointes	1 (0.1)
Ocular pathology	1 (0.1)
Skin disorder	1 (0.1)

MMSE=Mini-Mental State Examination; PET=Positron Emission Tomography; ARIA-

E=Amyloid-Related Imaging Abnormalities with Edema/Effusions;

AChEI=Acetylcholinesterase Inhibitor; MRI=Magnetic Resonance Imaging;

MHIS=Modified Hachinski Ischemia Scale; N/n=Number of participants.

Table S2. Characteristics of participants at baseline, including donanemab/BACE combo group

	Placebo (N=126)	Donanemab Monotherapy (N=131)	Donanemab / BACE Combo (N=15)	Total † (N=272)
Demographic				
Female sex, n (%)	65 (51.6)	68 (51.9)	12 (80.0)	145 (53.3)
Mean age, years (SD)	75.4 (5.4)	75.0 (5.6)	75.2 (6.0)	75.2 (5.5)
Race, n (%)				
Asian	2 (1.6)	1 (0.8)	0 (0)	3 (1.1)
Black or African American	3 (2.4)	5 (3.8)	0 (0)	8 (2.9)
White	121 (96.0)	122 (93.1)	15 (100.0)	258 (94.9)
Other*	0 (0)	3 (2.3)	0 (0)	3 (1.1)
Ethnicity, Hispanic/Latino [#] , n (%)	3 (2.4)	5 (3.8)	1 (6.7)	9 (3.3)
Education, ≥13 years, n (%)	102 (81.0)	97 (74.0)	10 (66.7)	209 (76.8)

APOE 4 carrier, n/N (%)	92/124 (74.2)	95/131 (72.5)	10/15 (66.7)	197/270 (73.0)
E2/E3, n (%)	1 (0.8)	1 (0.8)	0 (0)	2 (0.7)
E2/E4, n (%)	2 (1.6)	2 (1.5)	0 (0)	4 (1.5)
E3/E3, n (%)	31 (25.0)	35 (26.7)	5 (33.3)	71 (26.3)
E3/E4, n (%)	62 (50.0)	68 (51.9)	7 (46.7)	137 (50.7)
E4/E4, n (%)	28 (22.6)	25 (19.1)	3 (20.0)	56 (20.7)
AChEI use, n (%)	74 (58.7)	78 (59.5)	10 (66.7)	162 (59.6)
Scale, Mean (SD), range				
iADRS	105.9 (13.2), 67.0 – 139.0	106.2 (13.0) ^a , 60.0 – 130.0	108.9 (10.2), 88.0 – 123.0	106.2 (13.0) ^b , 60.0 – 139.0
CDR-SB	3.4 (1.7), 0.5 – 8.0	3.6 (2.1), 0.5 – 11.0	3.6 (2.1), 1.5 – 8.0	3.5 (1.9), 0.5 – 11.0
ADAS-Cog ₁₃	27.5 (7.6), 5.0 – 47.0	27.6 (7.7), 10.0 – 51.0	27.3 (7.3), 13.0 – 39.0	27.6 (7.6), 5.0 – 51.0
ADCS-ADL	67.0 (8.1), 40.0 – 78.0	67.4 (8.6) ^a , 28.0 – 78.0	69.9 (5.6), 56.0 – 78.0	67.3 (8.2) ^b , 28.0 – 78.0

ADCS-iADL	48.4 (7.5), 24.0 – 59.0	48.9 (7.6) ^a , 21.0 – 59.0	51.2 (5.5), 38.0 – 59.0	48.8 (7.5) ^b , 21.0 – 59.0
MMSE	23.7 (2.9) ^c , 16.0 – 29.0	23.6 (3.1) ^d , 14.0 – 29.0	21.5 (5.0) ^e , 13.0 – 30.0	23.5 (3.1) ^f , 13.0 – 30.0
Amyloid PET Centiloids, Mean (SD), range	101.1 (33.3), 38.7 – 225.2	107.6 (36.0), 41.0 – 251.4	99.2 (35.6), 42.1 – 167.1	104.2 (34.8), 38.7 – 251.4
Flortaucipir PET global tau load, Mean (SD), range	0.46 (0.15) ^g , 0.2 – 0.9	0.47 (0.19) ^h , 0.1 – 1.2	0.45 (0.15), 0.2 – 0.7	0.46 (0.17) ⁱ , 0.1 – 1.2

*Note: includes Multiple & American Indian or Alaska Native. † includes participants in the combo group. [#] number of participants with non-missing data, used as denominator, ^a Donanemab monotherapy N=130, ^b Total N=271, ^c Placebo N=121, ^d Donanemab monotherapy N=126, ^e Donanemab / BACE Combo N=14, ^f Total N=261, ^g Placebo N =124, ^h Donanemab monotherapy N =130, ⁱ Total N=269. APOE 4=Apolipoprotein E allele 4; AChEI=Acetylcholinesterase Inhibitor; ADAS-Cog₁₃=AD Assessment Scale-Cognitive 13-item Subscale; ADCS-ADL=Alzheimer's Disease Cooperative Study Activities of Daily Living Inventory; ADCS-iADL=Alzheimer's Disease Cooperative Study-instrumental Activities of Daily Living Inventory; iADRS=integrated Alzheimer's Disease Rating Scale; MMSE=Mini-Mental State Examination; CDR-SB=Clinical Dementia Rating Scale Sum of Boxes; PET=Positron Emission Tomography; N/n=Number of participants; SD=Standard Deviation.

Table S3. Primary iADRS, and secondary CDR-SB, ADAS-Cog₁₃, ADCS-iADL, and MMSE clinical outcomes

	iADRS	CDR-SB	ADAS-Cog₁₃	ADCS-iADL	MMSE
Week 12					
Placebo, LS mean change (SE)	-0.07 (0.684)	0.26 (0.126)	-0.00 (0.438)	-0.08 (0.455)	-0.68 (0.272)
Donanemab, LS mean change (SE)	0.42 (0.673)	0.21 (0.124)	-0.40 (0.433)	-0.02 (0.448)	-0.58 (0.267)
LS mean change difference (95% CI)	0.49 (-1.24, 2.22)	-0.05 (-0.37, 0.27)	-0.40 (-1.51, 0.71)	0.06 (-1.07, 1.20)	0.10 (-0.58, 0.78)
Week 24					
Placebo, LS mean change (SE)	-1.18 (0.666)	0.39 (0.123)	0.18 (0.451)	-0.91 (0.481)	-0.70 (0.295)
Donanemab, LS mean change (SE)	0.12 (0.663)	0.26 (0.123)	-0.33 (0.450)	-0.21 (0.478)	-0.71 (0.292)

LS mean change difference (95% CI)	1.30 (-0.38, 2.99)	-0.13 (-0.44, 0.18)	-0.51 (-1.66, 0.65)	0.69 (-0.52, 1.91)	-0.01 (-0.77, 0.74)
Week 36					
Placebo, LS mean change (SE)	-3.17 (0.730)	0.83 (0.139)	1.36 (0.445)	-1.73 (0.556)	-1.19 (0.314)
Donanemab, LS mean change (SE)	-0.73 (0.732)	0.26 (0.140)	0.53 (0.446)	-0.19 (0.558)	-0.89 (0.312)
LS mean change difference (95% CI)	2.44 (0.55, 4.33)	-0.56 (-0.92, -0.20)	-0.84 (-1.98, 0.30)	1.54 (0.09, 2.99)	0.30 (-0.51, 1.11)
Week 52					
Placebo, LS mean change (SE)	-6.70 (0.929)	1.21 (0.160)	2.37 (0.536)	-4.28 (0.635)	-1.56 (0.321)
Donanemab, LS mean change (SE)	-3.03 (0.933)	0.62 (0.160)	1.53 (0.540)	-1.64 (0.637)	-1.17 (0.321)
LS mean change difference (95% CI)	3.67 (1.19, 6.15)	-0.59 (-1.01, -0.16)	-0.84 (-2.25, 0.58)	2.64 (0.96, 4.33)	0.40 (-0.44, 1.23)

Week 64					
Placebo, LS mean change (SE)	-8.34 (1.038)	1.33 (0.171)	3.30 (0.621)	-4.91 (0.689)	-2.25 (0.339)
Donanemab, LS mean change (SE)	-4.92 (1.038)	1.06 (0.170)	1.87 (0.619)	-3.07 (0.687)	-1.34 (0.335)
LS mean change difference (95% CI)	3.42 (0.63, 6.21)	-0.27 (-0.72, 0.18)	-1.43 (-3.09, 0.23)	1.85 (0.01, 3.69)	0.91 (0.02, 1.79)
Week 76					
Placebo, LS mean change (SE)	-10.06 (1.141)	1.58 (0.178)	4.77 (0.660)	-5.20 (0.743)	-2.98 (0.390)
Donanemab, LS mean change (SE)	-6.86 (1.135)	1.22 (0.176)	2.91 (0.659)	-3.98 (0.738)	-2.35 (0.386)
LS mean change difference (95% CI)	3.20 (0.12, 6.27)	-0.36 (-0.83, 0.12)	-1.86 (-3.63, -0.09)	1.21 (-0.77, 3.20)	0.64 (-0.40, 1.67)

Results for the mean change from baseline for the primary iADRS, and secondary ADAS-Cog₁₃, ADCS-iADL, CDR-SB, and MMSE clinical outcomes, analyzed with MMRM. iADRS=integrated Alzheimer's Disease Rating Scale; ADAS-Cog₁₃=Alzheimer's Disease Assessment Scale-Cognitive subscale; ADCS-iADL=Alzheimer's Disease Cooperative Study-instrumental Activities of Daily Living scale; CDR-SB=Clinical Dementia Rating Scale Sum of Boxes; MMSE=Mini Mental State Examination; MMRM=Mixed Models for Repeated Measures; LS=Least Squares; CI=Confidence Interval; SE=Standard Error

Table S4. Summary of serious adverse events

	Placebo (N=125)	Donanemab (N=131)	p-value
Participants with ≥ 1 SAE, n (%)	22 (17.6)	23 (17.6)	>0.99
SAE, n (%)			
Pneumonia	1 (0.8)	5 (3.8)	0.21
Dehydration	1 (0.8)	2 (1.5)	>0.99
Hip fracture	2 (1.6)	1 (0.8)	0.62
Amyloid related imaging abnormality-edema/effusion	0 (0)	2 (1.5)	0.50
Infusion related reaction	0 (0)	2 (1.5)	0.50
Pulmonary embolism	0 (0)	2 (1.5)	0.50
Syncope	0 (0)	2 (1.5)	0.50
Acute myocardial infarction	1 (0.8)	0 (0)	0.49
Agitation	1 (0.8)	0 (0)	0.49
Anaemia	0 (0)	1 (0.8)	>0.99
Aortic aneurysm	1 (0.8)	0 (0)	0.49

Asthenia	1 (0.8)	0 (0)	0.49
Atrial fibrillation	0 (0)	1 (0.8)	>0.99
Atrioventricular block complete	1 (0.8)	0 (0)	0.49
Bacterial sepsis	0 (0)	1 (0.8)	>0.99
Bile duct stone	1 (0.8)	0 (0)	0.49
Bladder squamous cell carcinoma stage II	1 (0.8)	0 (0)	0.49
Breast cancer	0 (0)	1 (0.8)	>0.99
Brief psychotic disorder without marked stressors	1 (0.8)	0 (0)	0.49
Cardiac arrest	1 (0.8)	0 (0)	0.49
Cerebrovascular accident	0 (0)	1 (0.8)	>0.99
Clostridium difficile colitis	0 (0)	1 (0.8)	>0.99
Cognitive disorder	1 (0.8)	0 (0)	0.49
Confusional state	1 (0.8)	0 (0)	0.49
Diabetes mellitus	1 (0.8)	0 (0)	0.49

Dizziness	1 (0.8)	0 (0)	0.49
Drug eruption	0 (0)	1 (0.8)	>0.99
Encephalopathy	0 (0)	1 (0.8)	>0.99
Extradural haematoma	0 (0)	1 (0.8)	>0.99
Fall	1 (0.8)	0 (0)	0.49
Gastroenteritis	0 (0)	1 (0.8)	>0.99
Gout	1 (0.8)	0 (0)	0.49
Humerus fracture	1 (0.8)	0 (0)	0.49
Hypertension	1 (0.8)	0 (0)	0.49
Hypotension	1 (0.8)	0 (0)	0.49
Motor neurone disease	1 (0.8)	0 (0)	0.49
Mouth swelling	1 (0.8)	0 (0)	0.49
Non-cardiac chest pain	1 (0.8)	0 (0)	0.49
Non-small cell lung cancer stage II	0 (0)	1 (0.8)	>0.99
Orthostatic hypertension	0 (0)	1 (0.8)	>0.99

Osteoarthritis	1 (0.8)	0 (0)	0.49
Pelvic fracture	1 (0.8)	0 (0)	0.49
Pharyngeal swelling	1 (0.8)	0 (0)	0.49
Pneumonia aspiration	1 (0.8)	0 (0)	0.49
Psychotic disorder	1 (0.8)	0 (0)	0.49
Rheumatoid arthritis	0 (0)	1 (0.8)	>0.99
Road traffic accident	1 (0.8)	0 (0)	0.49
Seroma	1 (0.8)	0 (0)	0.49
Sinus node dysfunction	0 (0)	1 (0.8)	>0.99
Small intestinal obstruction	0 (0)	1 (0.8)	>0.99
Squamous cell carcinoma of skin	1 (0.8)	0 (0)	0.49
Staphylococcal infection	1 (0.8)	0 (0)	0.49
Subdural haematoma	0 (0)	1 (0.8)	>0.99
Upper limb fracture	1 (0.8)	0 (0)	0.49
Urinary tract infection	1 (0.8)	0 (0)	0.49

SAE=Serious Adverse Event; N/n = Number of participants

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